

Copula-Based Analysis of Flood Peak Level and Duration: Two Case Studies in Taihu Basin, China

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Abstract: Recently, several bivariate and multivariate methods have been used to analyze the relationship of flood peak discharges with corresponding hydrograph volumes and durations in many basins. In some flat tidal areas, such as the Taihu Basin of China, the flood peak level and the corresponding flood duration have received more attention than the flood peak discharge and other flood characteristics in flood control and flood management. In this study, avoiding the complex problem of positive-negative flows, a pragmatic approach for bivariate flood frequency analyses based on flood peak level data is proposed. Commonly used bivariate copulas and techniques for fitting such copulas are applied to analyze the correlation between flood peak levels and durations based on peak-over-threshold flood series in Pingwang and Wuzhen. The goodness-of-fit test shows that the extreme-value copulas are not suitable for these two cases. Along with the tail dependence analysis, the Student copula ($\nu = 5$) is chosen as the best-fitting model. A comparison between the joint return period (years) associated with the event ($H > h_T$ or $D > d_T$) obtained through the Student copula ($\nu = 5$) and other bivariate copulas demonstrates that ignoring the tail dependence results in a significant overestimate of flood risk. The joint return periods of arbitrary combinations of flood peak levels and flood durations are obtained through the final fitted copula. Furthermore, a comparison between the cumulative probability curves of flood peak levels obtained through different univariate and bivariate methods confirms the correctness of this work. DOI: [10.1061/\(ASCE\)HE.1943-5584.0001661](https://doi.org/10.1061/(ASCE)HE.1943-5584.0001661). © 2018 American Society of Civil Engineers.

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Introduction

Floods are one of the most common hazards on Earth. Several studies have concluded that climate change is increasing the risk of floods (Milly et al. 2002; Wilby et al. 2008; Schiermeier 2011; Hirabayashi et al. 2013). Thus, flood frequency analysis provides a critical basis for flood hazard analysis (Zhong et al. 2014) and the design of hydraulic structures such as levees, dams (Goodarzi et al. 2012; Requena et al. 2013), and culverts.

Univariate flood frequency analysis focusing on flood peak discharge has been widely conducted (Hamed and Rao 2000; Katz et al. 2002; Wardhana and Hadipriono 2003; Gado and Nguyen 2016). Most researchers have treated a flood event as a univariate event, assuming that the severity of a flood can be defined only by the flood peak discharge. However, the results of such analyses cannot provide a complete assessment of flood severity and

occurrence because a flood is an inherently multivariate random event characterized by a peak, volume, and duration (Adamson et al. 1999).

In recent years, many researchers have focused on bivariate and multivariate flood frequency analyses. Additionally, some researchers have used bivariate and multivariate probability distributions, such as bivariate normal distributions (Sackl and Bergmann 1987; Goel et al. 1998), bivariate extreme-value distributions (Yue et al. 1999), and multivariate Gaussian and exponential distributions (Krstanovic and Singh 1987; Singh and Singh 1991) to describe the joint distributions of flood peaks, volumes, and durations. Most of these studies have assumed that the flood variables have the same type of marginal probability distribution (Zhang and Singh 2006). However, flood variables are dependent and do not generally follow the same type of marginal distribution. Therefore, some researchers have derived multivariate flood frequency distributions using copulas to avoid the use of the aforementioned assumption (Nelsen 2006). The joint distributions of flood peaks, volumes, and durations have been derived using a variety of copula methods (Zhang and Singh 2006; Bačová Mitková and Halmová 2014; Yin et al. 2014; Sraj et al. 2015).

The aforementioned studies treated a flood as a function of two (peak and volume or peak and duration) or three variables (peak, volume, and duration). Both the two- and three-variable functions are based on river discharge data. This method can be applied to most river basins in the world where continuously observed discharge data are available. However, in some flat tidal river networks, this method may not be applicable for various reasons. First, because of the tidal actions associated with the downstream sea and the low-lying terrain of the area, a river flow may sometimes be reversed. Thus, observed river flows have both positive and negative values, and a reliable stage-discharge relationship curve may not be available. Second, because most flat tidal river networks are located over low-lying terrain and receive abundant rainfall, the flood peak level (H), rather than the flood peak

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discharge (Q), receives the most attention in flood control and flood management. Third, because of special geographic conditions, the backwater resulting from the downstream tides lengthens the flood recession time. As a result, the flood durations in these areas are longer than those in other areas. Obviously, a longer flood duration (D) implies a higher flood risk. Therefore, none of the aforementioned flood frequency analysis methods are applicable or suitable for these flat tidal river networks. Most areas containing these networks are densely populated and highly developed. Recent flood events have resulted in substantial damages and loss of human life. Such events include the Brisbane and Thailand floods in 2011, severe tropical storm Fitow in 2013, and Typhoons Sarika and Meranti in 2016. Additionally, the flood protection measures designed on the basis of traditional methods of univariate flood frequency analysis have been largely ineffective. Thus, a bivariate method of flood frequency analysis that considers both the flood peak level and duration in areas with flat tidal river networks is needed.

The present study proposes a pragmatic approach for bivariate flood frequency analysis based on flood peak level (or water level) data. Two case studies in the Taihu Basin were conducted using the proposed approach. This approach avoids using flood peak discharge data, thereby avoiding the complex problem of positive-negative flows. Furthermore, compared with the traditional univariate methods of water level analysis, the proposed bivariate method of frequency analysis considers more flood characteristics. These flood characteristics were considered by addressing the association between the flood peak level and flood duration. In addition, the peak-over-threshold (POT) sampling technique was used in the approach, which increased the sample size. The proposed pragmatic approach is easy to understand and applicable to flood-plain management in flat tidal areas similar to the Taihu Basin in China.

Study Area and Data

Study Area

The Taihu Basin is located in the delta region of the Yangtze River in Eastern China, and it encompasses a total area of 36,895 km². The location of the Taihu Basin is shown in Fig. 1. The Basin lies in a subtropical zone and has a monsoon climate with an average

annual precipitation of 1,177 mm, which is concentrated in the summer. There are low hills along the western edge of the basin, although low-lying areas cover approximately 80% of the basin, with elevations between 3 and 4 m above mean sea level. These elevations are 2–3 m lower than the highest water level at the mouth of the Yangtze River and 5–6 m lower than the highest tide in Hangzhou Bay. Because the area is very flat, has low flow velocities, and has a drainage system influenced by high tides, the area is very prone to fluvial flooding, storm surges, and internal floods caused by local heavy rainfalls. The flat areas in the eastern and central parts of the basin form a typical flat tidal river network (Zhao et al. 2011; Han et al. 2014). Because of the special geographic conditions and the influence of the backwater resulting from the downstream tides, flood durations in Taihu Basin are usually very long. For instance, during the basin-wide flood that occurred in 1991, the duration of the high water level, which is higher than the warning water level (3.5 m for Pingwang), in Pingwang was 88 days (Wu and Guan 1999). Moreover, during another basin-wide flood in 1999, the duration of the high water level in Pingwang was 104 days (Ou and Wu 2001). There is no doubt that both high water level and its long duration play important roles in the massive flood losses. Examining the joint and conditional return period of flood peak levels and durations will provide a substantial contribution to flood control or flood management in the Taihu Basin.

Data

In this study, the daily average water level data from Pingwang (1963–2012) and Wuzhen (1956–2012) stations were collected by the Taihu Basin Authority (TBA) of the Chinese Ministry of Water Resources. As is shown in Fig. 1, both Pingwang and Wuzhen are key gauging stations in the southeastern Taihu Basin, and they are located on the Taipu River and Lanxi River, respectively. Fig. 2(a) shows an example time series of hourly water level data at Pingwang during Typhoon Fitow in 2013. The water level varies based on twelve-hour periodic cycles. This period of fluctuation is the same as the tidal pattern in Hangzhou Bay and the East China Sea. Fig. 2(b) shows the corresponding average values of daily discharge. Twenty-seven positive values and four negative values can be seen in the graph of daily average discharge in August 2013 at Pingwang. Both Figs. 2(a and b) show that the Taipu River is a tidal river with a flow direction that is not fixed. The data from Wuzhen show that the flows in the Lanxi River exhibit a similar pattern as those in the Taipu River. Because long-term hourly water level data are unavailable, daily average water levels are used in this paper.

Methodology

In this paper, a flood event is described as a function of two variables: flood peak level (H) and flood duration (D). Flood event sample series are obtained from continuously observed water level data using the POT sampling technique after selecting optimal threshold levels. The series of flood peak level are fitted with generalized Pareto distributions (GPDs), whereas the flood duration series are fitted with other suitable distributions. The joint and conditional probability distributions of the flood peak level and flood duration are obtained after selecting the most appropriate bivariate copula through a goodness-of-fit test and a tail dependence analysis. Different copulas are then used to estimate the joint return periods (years) associated with events ($H > h_T$ or $D > d_T$). The conditional probability distribution function is derived and used to estimate the cumulative probability of a given flood peak level for different flood durations.

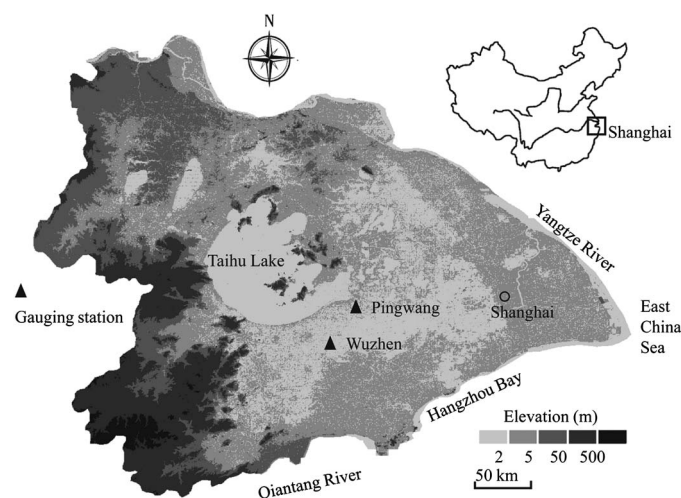


Fig. 1. Topography of the Taihu Basin and locations of the two gauging stations (Pingwang and Wuzhen).

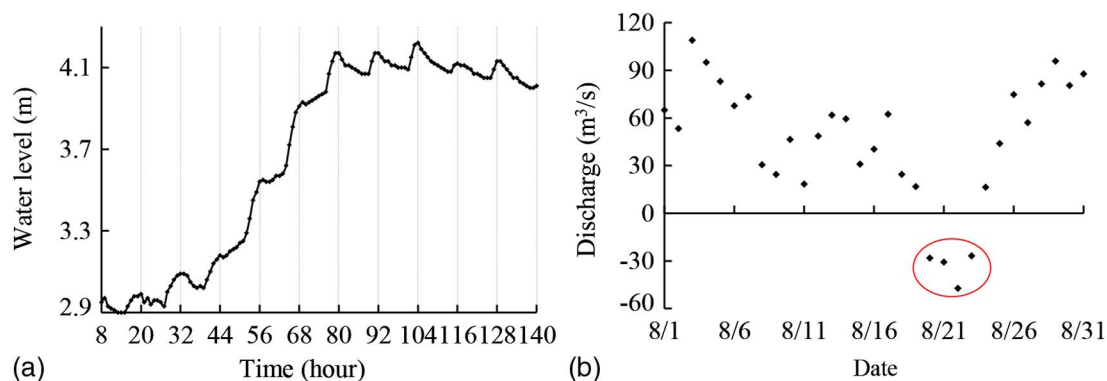


Fig. 2. Water level and discharge in a tidal river: (a) hourly water level; and (b) daily average discharge.

Sampling Technique for POT Series

The annual maximum series (AMS) and POT approaches are regarded as the most common flood sampling methods for flood frequency analysis (Lang et al. 1999; Zhou et al. 2014; Gaál et al. 2015). The AMS method consists of only the maximum value in each year, whereas the POT series consists of all peak values that exceed a certain threshold. The main advantage of POT sampling is that it allows for a more rational selection of events that are considered flood events. Compared with the AMS approach, POT sampling provides additional flexibility in the representation of floods and a more complete description of the flood generation process, because the observed hydrologic time series are always very short. In this paper, the POT sampling approach is used.

Generalized Pareto Distribution

In recent years, many researchers have suggested that the two-parameter GPD is flexible and fits many POT series well (Li et al. 2014). The cumulative probability distribution function of flood peak level (H) with a GPD (Pickands 1975) distribution can be written as follows:

$$F_H(h) = 1 - \left\{ 1 + \left[\frac{k(h - \mu)}{\alpha} \right] \right\}^{-(1/k)} \quad (1)$$

where μ , α , and k = location, scale, and shape parameters of the GPD, respectively. The variable μ (or threshold) must be specified before fitting Eq. (1) because the GPD is a distribution of threshold excesses.

Threshold Selection for Generalized Pareto Distributions

Selecting an appropriate threshold is critical in GPD analysis. The threshold should be set both high enough so that the exceedances approximate Poisson arrivals and low enough to minimize the estimation variance when fitting the GPD to observed samples, which ensures that enough data points are available to determine the GPD parameters satisfactorily. Researchers have developed several graphical and numerical methods to assist in the selection of an optimal threshold (Langbein 1949; Madsen et al. 1993; Lang et al. 1999; Brabson and Palutikof 2000; Veneziano et al. 2009; Bezak et al. 2014; Li et al. 2014; Sherly et al. 2016). However, determining the most effective methods remains a topic of debate. Here, a graphical method (Brabson and Palutikof 2000; Sherly et al. 2016) is used to obtain the optimal threshold. This method works as follows: (1) the shape and the scale parameters of the fitted GPDs are plotted against increasing threshold values, and (2) the lowest threshold at which the parameters start to become reasonably stable (i.e., parameter values do not change significantly with small changes in the threshold) is chosen as the optimal threshold. To

confirm that the thresholds are optimal, two additional numerical methods (Langbein 1949; Madsen et al. 1993) are also used and discussed.

Independence Criterion

The individual observations comprising a POT series for GPD fitting should be independent of each other. To verify whether this independence condition is satisfied, several criteria have been proposed (Cunnane 1979; Hydrology Subcommittee 1982). The selection of an appropriate independence criterion is a complex problem. In this paper, on the basis of previous studies, an independence criterion that is suitable for the study area is proposed. The concentration time T_c is used to separate successive flood events; thus, the interval time between two flood events must be longer than the concentration time. Otherwise, the two flood events would be regarded as one flood event. This independence criterion is illustrated in Fig. 3. The runoff concentration time in the flat areas of the Taihu Basin is approximately 3 days (Cheng et al. 2006). As a result, T_c of 3 days is used to separate successive flood events. In addition, before any frequency analysis, the independence assumption for the flood variables should be tested. In this paper, a lag-autocorrelation test is used to evaluate the independence assumption. If the test fails, the threshold selected in the previous section should be revised.

Bivariate Frequency Analysis

Copula Method

Hydrological phenomena are often multidimensional and require the joint modeling of several random variables (Genest and Favre 2007). Traditional methods of multivariate frequency analysis

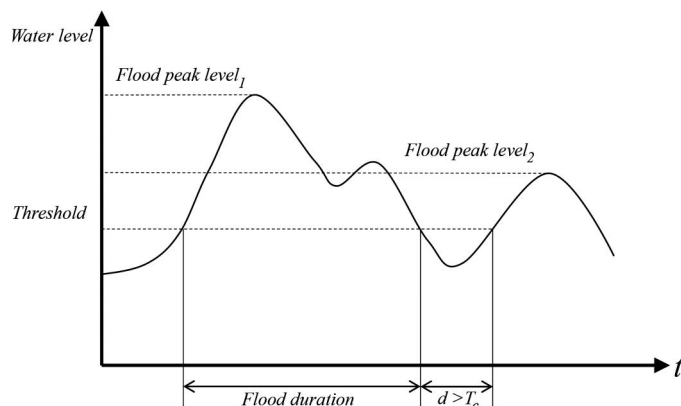


Fig. 3. Independence criterion for flood event sampling.

assume that marginal distributions are from the same family of distributions and the dependence between variables follows a linear relationship (Requena et al. 2013). However, limitations arise because these assumptions cannot be satisfied by the dependence structure of flood variables. Thus, the copula method can be used to address related difficulties. The copula method is based on Sklar's theorem (Sklar 1959). According to Sklar's theorem, if M is a joint cumulative distribution function with marginal distribution functions F_H and F_D , then a copula C exists, such that all h and $d \in R$

$$M(h, d) = C\{F_H(h), F_D(d)\} \quad (2)$$

For additional details regarding the copula method, see Nelsen (2006). The definitions and parameter domains of the seven copulas used in this paper are given in the Appendix.

Dependence Evaluation and Parameter Estimation

A dependence analysis between correlated random variables is conducted to determine if some type of dependence can be deduced from the data (Nelsen 2006). It can be performed via graphical analysis, such as using a chi-plot (Fisher and Switzer 1985, 2001) or K-plot (Genest and Boies 2003), or via dependence measures such as Spearman's ρ or Kendall's τ (Lehmann 1966).

The estimation of the parameter θ of a copula family $C_\theta(u, v)$ that best fits the data can be obtained using different parametric and semiparametric methods, such as Kendall's τ method, Spearman's ρ method, and maximum likelihood-based methods (e.g., full maximum likelihood method, inference for margins method, and canonical maximum likelihood method) (Joe 1997, 2005). In this paper, seven commonly used one-parametric bivariate copulas are considered: Clayton, Frank, Gumbel, Galambos, Hüsler-Reiss, Normal, and Student. Different methods of parameter estimation are used for different families of copulas. For the first five copulas, the full maximum likelihood method (FML) is used. For the Gaussian copula and Student copula, Kendall's τ method is used because it is the most direct method for elliptic copulas, whereas the number of degrees of freedom of the Student copula is selected on the basis of a visual fit (Poulin et al. 2007).

Goodness-of-Fit Tests

The aim of a goodness-of-fit test is to select the copula that best represents the dependence structure of observed variables. Graphical tools and formal tests can be used to achieve this objective. Graphical tools can provide a general idea of the goodness of fit, whereas formal tests can quantify the fit. Three types of formal test procedures have been proposed: tests based on empirical process, tests based on Kendall's transform, and tests based on Rosenblatt's transformation (Rosenblatt 1952; Dobrić and Schmid 2007; Genest and Favre 2007; Berg 2009). Genest et al. (2009) presented a critical review of these goodness-of-fit procedures and proposed new techniques. They found that because of the arguable assumption that the limiting distribution is a chi-square distribution, the p-value calculation of some procedures might be incorrect. In this current paper, two Cramér-von Mises statistics S_n and $S_n^{(B)}$ are used because of their excellent power properties against a variety of alternatives (Genest et al. 2006, 2009). For the goodness-of-fit test for the last two extreme-value copulas (i.e., Galambos and Hüsler-Reiss), only the statistic S_n is used. The details of the functions of the statistics and the bootstrap methodology required to compute associated p-values are presented in Genest et al. (2009, 2011). In the bootstrap, $N = 1,000$ random samples of size $n = N_s$ are simulated, and the copula parameters are estimated via the maximum likelihood method.

Tail Dependence Analysis

Tail dependence analysis is used to evaluate the copula behavior for high return periods. Poulin et al. (2007) highlighted the importance of considering the tail dependence in bivariate frequency analysis based on copulas. In this study, more attention is paid to the upper-tail dependence because the focus of this study is on the frequency analysis of extreme flood events for flood control. Upper-tail dependence is associated with the capacity to link extreme flood peak levels (H) to extreme flood durations (D). The upper-tail dependences are specific in the copula model and its parameters. The values of the upper-tail dependence coefficients λ_U for Gumbel and Student copulas could be calculated using the following formulas $2 - 2^{1/\theta}$ and $2t_{\nu+1}(-\sqrt{(\nu+1)(1-\theta)/(1+\theta)})$, respectively, whereas the values of λ_U for some of the other copulas (e.g., Clayton, Frank, and Normal) are equal to zero.

The aim of this section is to identify the copula models that suitably reproduce the dependence of the extremes. Thus, the upper-tail dependence coefficient obtained from the copula models should be compared with the coefficient estimated from the observed data. Three nonparametric estimators of the upper-tail dependence coefficient are commonly used, i.e., $\hat{\lambda}_U^{\text{LOG}}$, $\hat{\lambda}_U^{\text{SEC}}$, and λ_U^{CFG} . The first two estimators ($\hat{\lambda}_U^{\text{LOG}}$ and $\hat{\lambda}_U^{\text{SEC}}$) are defined by using a threshold. On the basis of the property of homogeneity of the tail copula, Schmidt and Stadtmüller (2006) proposed an iterative algorithm to choose the threshold. In this study, as advocated by Frahm et al. (2005), a bandwidth $b = \lfloor 0.005n \rfloor = 0$ for the $\hat{\lambda}_U^{\text{LOG}}$ and $\hat{\lambda}_U^{\text{SEC}}$ estimators is assumed such that each moving average contains 1% of the data. In the case of $\hat{\lambda}_U^{\text{LOG}}$ estimators, a larger bandwidth equal to 1 is also assumed to reduce the variance of the estimations. The third estimator λ_U^{CFG} has the advantage that no threshold is necessary. However, this estimator is based on the assumption that the empirical copula can be approximated by an extreme-value copula. Serinaldi et al. (2015) showed that this assumption is not weak and that λ_U^{CFG} systematically points to extreme-value copulas. In this study, because of the results of the goodness-of-fit test, which are presented in the "Copula Selection" section, the underlying assumption of the λ_U^{CFG} estimator will be invalid. Although $\hat{\lambda}_U^{\text{LOG}}$ and $\hat{\lambda}_U^{\text{SEC}}$ are asymptotically equivalent because of the local behavior of the logarithm function near 1, Poulin et al. (2007) indicated that it is useful to consider both of them, especially when dealing with small samples. The functions of the estimators can be found in many published articles (Frahm et al. 2005; Poulin et al. 2007); they are not repeated herein for conciseness.

Joint and Conditional Return Period

In hydrological planning and design, it is important to estimate the return period because, for a given flood event period, it is possible to obtain various occurrence combinations of H and D (Vittal et al. 2015). The joint return period represents an event in which h , d , or both are exceeded [$(H > h \text{ or } D > d) \text{ or } (H > h \text{ and } D > d)$]. In this study, only the scenario in which either h or d is exceeded is analyzed. The joint return period (recurrence interval) of a flood event with a flood peak level (H) and flood duration (D) from a POT flood event series can be calculated as follows:

$$T_{H,D} = \frac{N_e}{N_s P(H > h \cup D > d)} = \frac{N_e}{N_s (1 - M_{H,D})} \quad (3)$$

where N_e = total length of the flood time series (years); N_s = total number of sampled extremes used for the calibration of a theoretical distribution; and $M_{H,D}$ = joint cumulative probability of H and D [i.e., $M_{H,D} = P(H \leq h, D \leq d)$]. For any values of h and d , $M_{H,D}$ can be calculated by using the fitted joint distribution C

of random variables H and D . Then, the conditional return period of event $H > h$ when $D = d$ is expressed as follows:

$$T(h|d) = \frac{N_e}{N_s(1 - M(h|d))} \quad (4)$$

Results and Discussion

Peak-over-Threshold Series

A graphical method was initially used to obtain the optimal thresholds at which the scale and shape parameters of the GPD are relatively stable. The shape parameter k and scale parameter α of the GPD are plotted against varying thresholds μ in Fig. 4. Fig. 4 shows that the optimal thresholds for Pingwang and Wuzhen are 3.35 and 3.53, respectively. Because the thresholds are key parameters in this study, two additional numerical methods to estimate optimal values are also explored. First, as suggested by Langbein (1949), the threshold can be selected as equal to the lowest annual maximum water level. This means that at least one event per year is included in the analysis. As a result, 2.83 and 3.03 are selected as the possible thresholds at Pingwang and Wuzhen, respectively. The two water levels are obviously inappropriate because they are even lower than their nearby ground levels. Second, the standard frequency factor k_0 and data properties (mean value μ_x and standard deviation σ_x) can be used to calculate the threshold value μ using the expression $\mu = \mu_x + k_0\sigma_x$ (Madsen et al. 1993). The standard frequency factor $k_0 = 1.5$ is selected. Because the mean values are 2.92 and 3.03, and the standard deviations are 0.30 and 0.35, 3.37 and 3.56 are obtained as thresholds for Pingwang and Wuzhen, respectively.

Because the optimal thresholds obtained through the graphical and numerical methods are nearly equal, both methods are found to be valid. The results obtained through the graphical method are used subsequently in the study. Furthermore, a lag-autocorrelation test is used to test the independence of the events comprising the resulting flood peak level series and the corresponding flood duration series. The result indicates that both the series of flood peak

level and the corresponding flood duration series at the two stations are independent at the 0.05 significance level because the autocorrelation coefficients of different time lags are within the corresponding critical values. Therefore, the thresholds are confirmed as the optimal thresholds, and the POT series can be obtained. In total, 105 and 111 flood event samples are obtained from 50- and 57-year water level time series at Pingwang and Wuzhen, respectively. A large sample size reduces the uncertainty in the frequency analysis results. The results of the lag-autocorrelation test are shown in Table 1. Because all the tests were passed, the POT series obtained under the previous threshold are used subsequently in the study.

Marginal Frequency of the Flood Peak Level

First, the series of flood peak level are fitted to GPDs. The goodness of fit of each GPD is tested using the Kolmogorov-Smirnov (K-S) test and chi-square test. Regarding the chi-square test, the number of degrees of freedom is calculated as $r - c - 1$, where r is the number of classes, and c is the number of parameters to be estimated. The desirable number of classes is estimated by using the expression $5\log_{10}(N_s)$, where N_s is the total number of sampled extremes (Mutua 1994). Table 2 lists the calculated test statistics. For comparison, other commonly used models, such as log-Pearson Type III (LP3) and 3P-lognormal (3PLN) models, are also tested for fitting the series of flood peak level in this study. The results of the goodness-of-fit tests are also shown in Table 2. Table 2 shows that the p-values for the GPD are larger than those for the other models, indicating that the GPD provides the best fit at both Pingwang and Wuzhen. Fig. 5 also shows the goodness-of-fit of the GPD.

Marginal Frequency of the Flood Duration

The flood duration series are fitted using different models. The goodness-of-fit values of the four models are quantified by using the K-S and chi-square test statistics. Table 2 shows that the fits of all three models are good and that their differences are very small. For convenience, we recommend the lognormal model to fit the

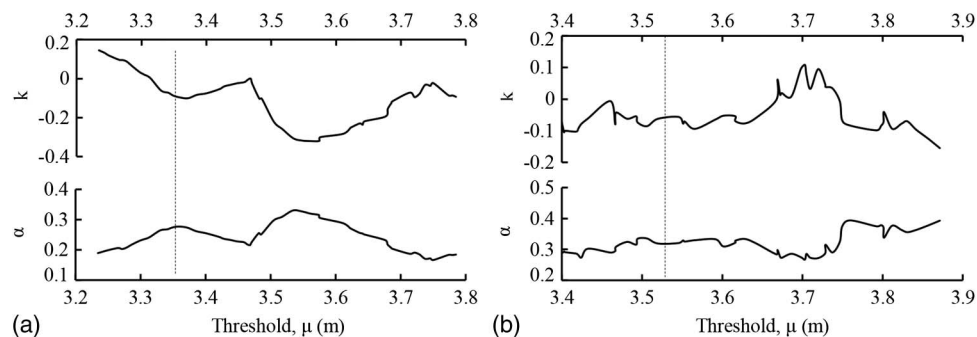


Fig. 4. Graphical method of optimal threshold selection based on scale parameter α versus threshold μ and shape parameter k versus threshold μ for (a) Pingwang; and (b) Wuzhen.

Table 1. Results of lag-autocorrelation test

Station	Lag	1	2	3	4	5
Pingwang	H	Pass (0.36)	Pass (0.11)	Pass (0.21)	Pass (0.28)	Pass (0.40)
	D	Pass (0.96)	Pass (0.47)	Pass (0.68)	Pass (0.81)	Pass (0.86)
Wuzhen	H	Pass (0.24)	Pass (0.50)	Pass (0.70)	Pass (0.15)	Pass (0.22)
	D	Pass (0.29)	Pass (0.58)	Pass (0.75)	Pass (0.55)	Pass (0.20)

Note: Pass or fail at a p -value level of significance = 0.05.

Table 2. Results of goodness-of-fit tests for different distributions of flood peak levels and flood durations

Variable	Distribution	K-S		Chi-square (6 degrees of freedom)	
		Pingwang	Wuzhen	Pingwang	Wuzhen
<i>H</i>	GPD	Pass (0.60)	Pass (0.52)	Pass (0.63)	Pass (0.93)
	LP3	Pass (0.10)	Pass (0.06)	Pass (0.28)	Pass (0.90)
	3PLN	Pass (0.27)	Pass (0.30)	Pass (0.27)	Pass (0.38)
<i>D</i>	Burr	Pass (0.63)	Pass (0.68)	Pass (0.30)	Pass (0.29)
	Log-logistic	Pass (0.68)	Pass (0.49)	Pass (0.28)	Pass (0.22)
	Lognormal	Pass (0.75)	Pass (0.50)	Pass (0.75)	Pass (0.40)

Note: Pass or fail at a *p*-value level of significance = 0.05.

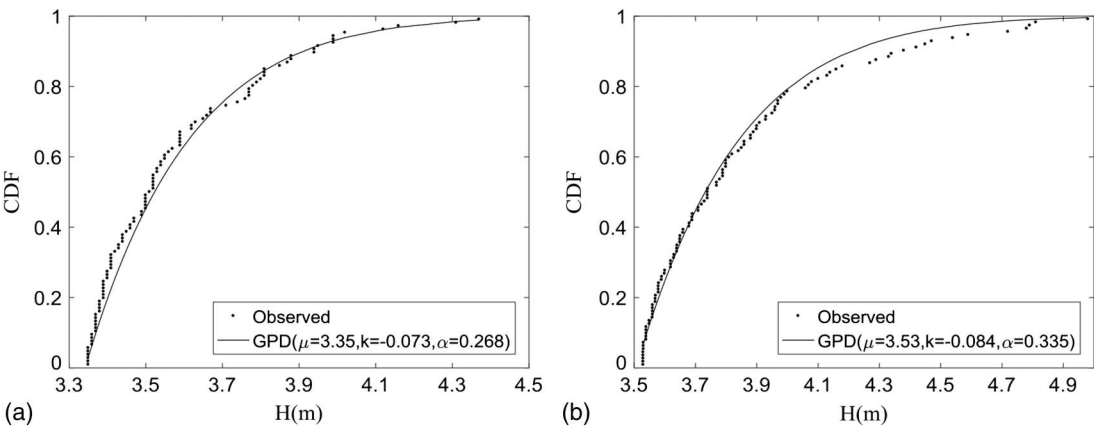


Fig. 5. Flood peak level fitted using the GPD at (a) Pingwang; and (b) Wuzhen. CDF = cumulative distribution function.

flood duration series at both Pingwang and Wuzhen. The cumulative probability distribution of the lognormal model is as follows (Cox 1946):

$$F_D(d) = \Phi\left(\frac{\log(d) - \mu_d}{\sigma_d}\right) \tag{5}$$

where Φ = Laplace integral; and σ_d and μ_d = location and scale parameters, respectively.

Copula Selection

Using Kendall’s τ method and the FML method, the parameters of the seven candidate copulas are estimated. For the Student copula, $\theta = \sin[(\pi/2)\tau]$, and the number of degrees of freedom is selected via visual fitting. By superimposing the observed couples on synthetic ν , Fig. 6 shows that, at both Pingwang and Wuzhen, the best fit is achieved for $\nu = 5$. To evaluate the performance of the models quantitatively, tests based on the statistics S_n and $S_n^{(B)}$ are used to check the goodness-of-fit of the seven copula models. Table 3 shows the estimated parameters of the copulas and the corresponding results of the goodness-of-fit tests. We assume that the goodness-of-fit performance of the copula is considered good only if both tests pass (i.e., *p*-values larger than the significance level of 0.05). Thus, the goodness-of-fit performance of the Clayton and Student copulas is considered good at Pingwang. At Wuzhen, the following three copulas performs well: Clayton, Normal, and Student. Given the poor performance of the three extreme-value copulas (i.e., Gumbel, Galambos, and Hüsler-Reiss), there is no good reason to assume that the copula could be approximated by an extreme-value copula. Consequently, λ_U^{CFG} is not used in the tail dependence analysis.

To evaluate the behaviors of the candidate copulas for high return periods. The upper-tail dependence coefficients estimated from sample series and computed from the formulas are compared. The upper-tail dependence coefficients of Clayton and Normal are equal to zero, which suggests that those two copulas cannot describe the tail dependence of the sample series. Table 4 shows that the results obtained through the three empirical estimators (i.e., λ_U^{LOG} with $b = 0$, λ_U^{LOG} with $b = 1$, and λ_U^{SEC}) are similar. Notably, among the candidate copulas passing the goodness-of-fit test, the upper-tail dependence coefficient of the Student copula ($\nu = 5$) is closest to the empirical values. As a result, the Student copula ($\nu = 5$) is used to model the bivariate distribution of the correlated flood peak levels and durations at both Pingwang and Wuzhen. Ten thousand *H*-*D* pairs are randomly generated by using the selected copula. Comparisons between these *H*-*D* pairs and the observed sample series are displayed in Figs. 6(a–f) for Pingwang and Figs. 6(g–l) for Wuzhen. The results confirm that the Student copula ($\nu = 5$) represents the dependence structure between the flood peak level and the flood duration. Thus, it can be applied to model the flood characteristics at Pingwang and Wuzhen.

Joint and Conditional Return Periods

As explained in the previous subsection, the Student copula ($\nu = 5$) is selected as the most appropriate model for modeling the correlated flood peak level and flood duration pairs at both Pingwang and Wuzhen. Using Eq. (3), the joint return periods of arbitrary combinations of *H* and *D* can be obtained (as shown in Fig. 7). For example, the joint flood event with $H > 4.5$ m or $D > 20$ days has a return period of 2.53 years at Pingwang; for $H > 5$ m or $D > 100$ days, the return period is 53.44 years. Similarly, at Wuzhen, the joint flood event with $H > 4.5$ m or $D > 20$ days

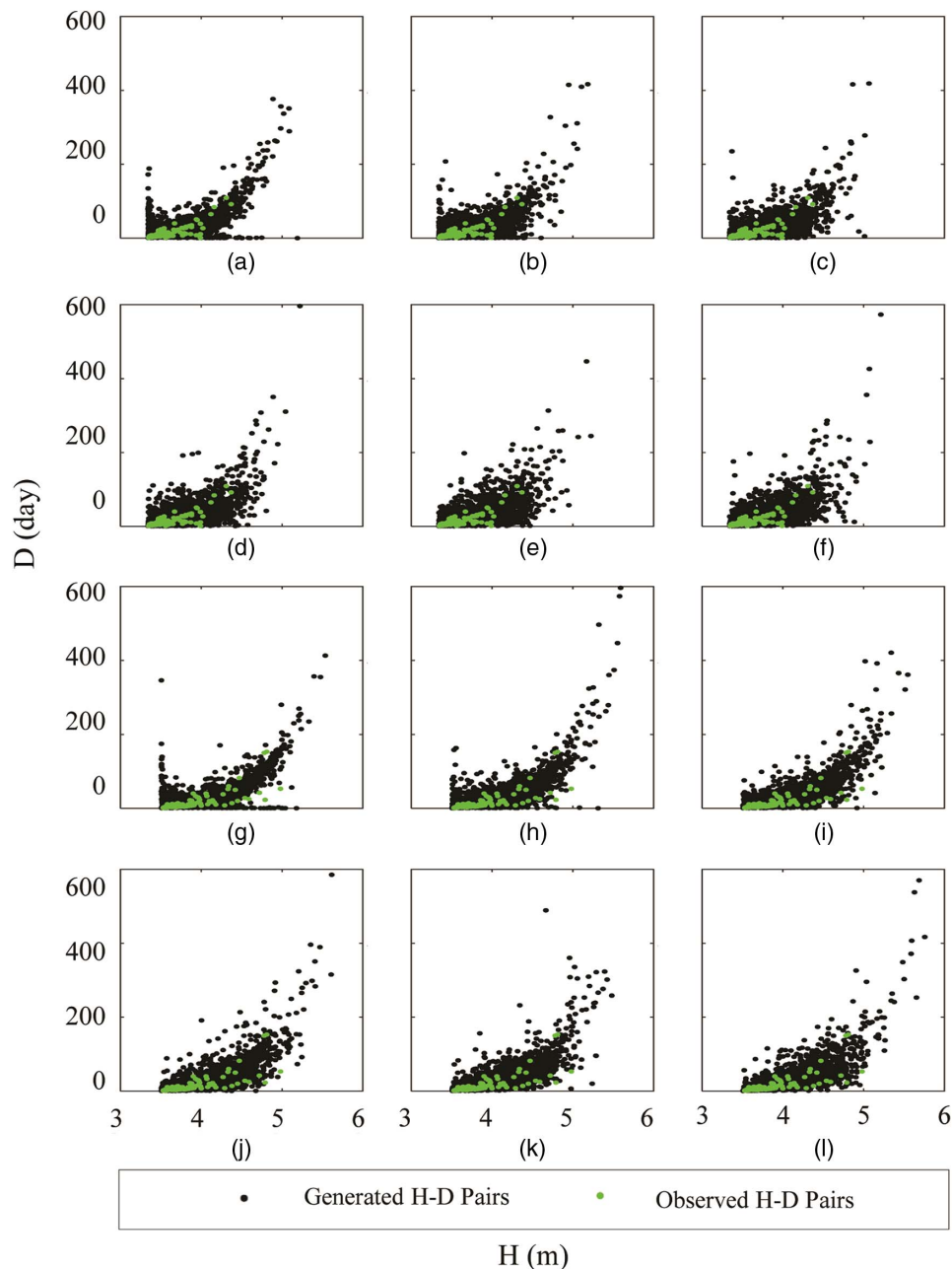


Fig. 6. Simulated random sample size of 10,000 from a Student copula: (a–f) Pingwang; and (g–l) Wuzhen. The degree of freedom ν considered ranges from 1 to 6: (a and g) $\nu = 1$; (b and h) $\nu = 2$; (c and i) $\nu = 3$; (d and j) $\nu = 4$; (e and k) $\nu = 5$; and (f and l) $\nu = 6$.

has a return period of 2.65 years, and an event with $H > 5$ m or $D > 100$ days has a return period of 40.29 years.

Table 5 presents the joint return period (years) associated with an event ($H > h_T$ or $D > d_T$) for $T = 10, 20, 50$, and 100 years for the copulas that pass the goodness-of-fit tests. Compared with the Student copula ($\nu = 5$), the Clayton and Normal copulas (for Pingwang, only Clayton) underestimate the joint return period and significantly overestimate the flood risk. This characteristic is even more visible for large return periods ($T = 50$ and 100 years). Recall that these two copulas have an upper-tail dependence coefficient equal to zero, which means that they are unable to connect extreme flood peak levels to extreme durations.

The conditional joint distribution based on the copula method can now be expressed. An equivalent formula for the conditional cumulative density function of H when $D = d$ can be expressed as follows:

$$M(H|D = d) = C_{U|V=v}(u, v) = C(U \leq u|V = v) = \frac{\partial C(u, v)}{\partial v} \quad (6)$$

where $u = F_H(h)$; and $v = F_D(d)$. Then, by incorporating Eq. (4), the conditional return period of event $H > h$ when $D = d$ can be expressed as follows:

$$T(h|d) = \frac{N_e}{N_s(1 - \frac{\partial C(u, v)}{\partial v})} \quad (7)$$

As is shown in Fig. 8, the conditional cumulative frequency curves of the flood peak level are computed for durations of 10, 20, 50, and 100 days. For example, when the duration is equal to 20 days, the 20-year flood peak level is 4.44 m at Pingwang and 4.82 m at Wuzhen. Fig. 8 indicates that, as the duration

Table 3. Estimated parameters and the results of goodness-of-fit tests

Copula	Pingwang			Wuzhen		
	$\hat{\theta}$	S_n (p-value)	$S_n^{(B)}$ (p-value)	$\hat{\theta}$	S_n (p-value)	$S_n^{(B)}$ (p-value)
Clayton	2.65	0.116 (0.05)	0.063 (0.25)	3.18	0.150 (0.05)	0.092 (0.05)
Frank	8.17	0.108 (0.01)	0.085 (0.02)	10.85	0.088 (0.02)	0.072 (0.06)
Gumbel	2.60	0.108 (0.04)	0.112 (0.01)	2.91	0.123 (0.00)	0.105 (0.00)
Galambos	1.81	0.057 (0.01)	—	2.21	0.023 (0.04)	—
Hüsler-Reiss	2.36	0.007 (0.01)	—	2.87	0.025 (0.04)	—
Normal	0.84	0.099 (0.02)	0.127 (0.00)	0.87	0.071 (0.20)	0.043 (0.53)
Student ($\nu = 5$)	0.84	0.066 (0.75)	0.102 (0.05)	0.87	0.088 (0.05)	0.041 (0.62)

Note: The bold numbers represent successful tests because the p -value is larger than 0.05.

Table 4. Upper-tail dependence coefficients

Copula	$\hat{\lambda}_U$	
	Pingwang	Wuzhen
$\hat{\lambda}_U^{\text{LOG}} b = 0$	0.646	0.658
$\hat{\lambda}_U^{\text{LOG}} b = 1$	0.643	0.656
$\hat{\lambda}_U^{\text{SEC}}$	0.683	0.689
Clayton	0	0
Normal	0	0
Student ($\nu = 5$)	0.644	0.691

increases, the flood peak level within the same conditional return period increases dramatically. Fig. 8 also presents the cumulative frequency curve of the flood peak level obtained through the univariate GPD. The designed water levels of frequently occurring, short-duration ($D = 5$ days) floods obtained through the copula method are significantly lower than those obtained through the univariate GPD. This finding suggests that the designed water levels estimated by using traditional univariate methods might be conservative. Fig. 8 also presents the cumulative frequency curve of the flood peak level obtained through the univariate Pearson Type III (P3) based on AMS. According to SL44-2006 (Changjiang Water Resources Commission 2006), a univariate P3 distribution with AMS is the most commonly used method in the field of flood frequency analysis in China. The small difference between the curves obtained through the two univariate methods confirms that the GDP model used in this study is correct and accurate. It should be stressed that traditional univariate methods can provide only limited information regarding designed water levels.

Summary and Conclusion

In this paper, a bivariate flood frequency analysis is based on POT data series and bivariate copulas. After an initial POT analysis, a flood event is described as a function of two variables: flood peak level and flood duration. The independence of both the flood peak level and the flood duration series is evaluated by using a lag-autocorrelation test. The flood peak level series are fitted closely by GPDs, whereas the flood duration series are fitted by using different models. The joint and conditional probability distributions of the flood peak level and the flood duration are obtained after selecting the most appropriate bivariate copula. The conditional probability values are then used to estimate the cumulative probability of a flood peak level for a duration equal to a given value.

- The following conclusions are drawn from this study:
1. Using POT analysis, as many as 105 and 111 flood event samples are obtained from 50- and 57-year series of water level data at Pingwang and Wuzhen, respectively. Increasing the sample size can significantly reduce the uncertainty in the frequency analysis results.
 2. The Student copula ($\nu = 5$) is the best-fitting copula for bivariate flood variables at both Pingwang and Wuzhen, and the poor performance of the three extreme-value copulas indicates that such copulas are not applicable to this study.
 3. A comparison between the joint return period (years) associated with an event ($H > h_T$ or $D > d_T$) obtained through the Student copula ($\nu = 5$) and other bivariate copulas demonstrates that ignoring the tail dependence leads to incorrect results.
 4. Compared with the traditional univariate methods of flood frequency analysis, the copula-based approach considers both flood durations and flood peak levels, and the associations

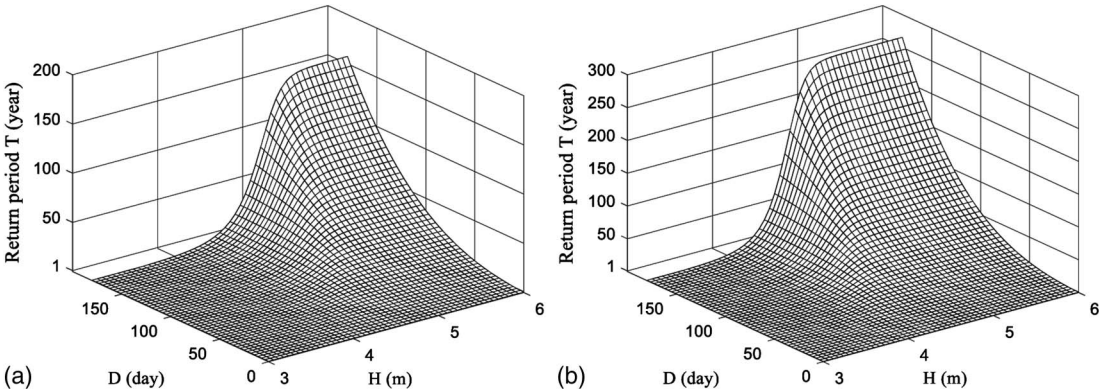
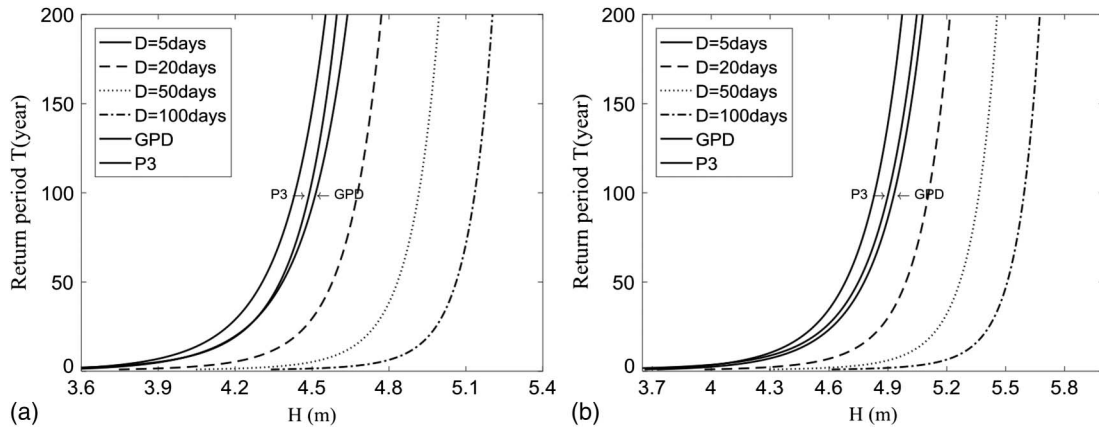


Fig. 7. Joint return periods of arbitrary combinations of H and D : (a) Pingwang; and (b) Wuzhen.

Table 5. Joint return period (years) associated with an event ($H > h_T$ or $D > d_T$) for $T = 10, 20, 50$, and 100 days

Copula	$T = 10$		$T = 20$		$T = 50$		$T = 100$	
	Pingwang	Wuzhen	Pingwang	Wuzhen	Pingwang	Wuzhen	Pingwang	Wuzhen
Clayton	5.84	5.51	11.24	10.52	27.42	25.53	54.39	50.53
Normal	—	6.88	—	13.29	—	31.96	—	62.38
Student	7.69	7.14	15.09	14.02	37.11	34.45	73.57	68.28

**Fig. 8.** Comparison between designed water levels obtained through the conditional return period and marginal GPD for (a) Pingwang; and (b) Wuzhen.

between these flood characteristics can be analyzed through dependence evaluations and properly represented by copula functions.

- The relation between flood peak level and flood duration can be investigated by avoiding the complex plus/minus streamflow problem that is common in tidal rivers (low-lying land). This pragmatic approach for bivariate flood frequency analysis is easy to understand and applicable to flat tidal areas similar to the Taihu Basin in China, where continuously observed discharge data are not available or exceedingly complex to analyze.

Further investigations and extending the present work are key. For example, in this study, because of the lack of necessary data (i.e., upstream rainfall and downstream tide data), the flood events are not classified according to their causes. Because different categories of flood events may have different characteristics (Szolgay et al. 2016), it will be beneficial in future work if the flood events are first classified before being analyzed by using copulas. Moreover, only parametric approaches are applied in this study, and there remains the opportunity of applying nonparametric approaches. Nonparametric methods can reduce potential systematic biases, which are common in parametric models (Sherly et al. 2016). Further investigation and extension would make this approach more applicable and useful.

Appendix. Copula Functions

The equations of several copulas used in this study are as follows:

- Archimedean copula

- Clayton copula

$$C(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}$$

where θ = parameter of copulas, $\theta \geq -1$ and $\theta \neq 0$.

- Frank copula

$$-\frac{1}{\theta} \ln \left[1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1} \right]$$

where θ = parameter of copulas, $\theta \neq 0$.

- Gumbel copula

$$\exp\{-[(-\ln u)^\theta + (-\ln v)^\theta]^{1/\theta}\}$$

where θ = parameter of copulas, $\theta \geq 1$.

- Extreme-value copula

- Galambos copula

$$uv \exp\{((- \ln u)^{-\theta} + (- \ln v)^{-\theta})^{-1/\theta}\}$$

where θ = parameter of copulas, $\theta \geq 0$.

- Hüsler-Reiss copula

$$\exp \left[-\tilde{u} \Phi \left\{ \frac{1}{\theta} + \frac{2}{\theta} \ln \left(\frac{\tilde{u}}{v} \right) \right\} - \tilde{v} \Phi \left\{ \frac{1}{\theta} + \frac{2}{\theta} \ln \left(\frac{\tilde{v}}{u} \right) \right\} \right]$$

where θ = parameter of copulas, $\theta \geq 0$; $\tilde{u} = -\ln u$; and $\tilde{v} = -\ln v$.

- Elliptical copula

- Normal copula

$$\int_{-\infty}^{\Phi^{-1}(u)} \int_{-\infty}^{\Phi^{-1}(v)} \frac{1}{2\pi\sqrt{1-\theta^2}} \exp \left(\frac{2\theta s\omega - s^2 - \omega^2}{2(1-\theta^2)} \right) ds d\omega$$

where θ = parameter of copulas, $-1 \leq \theta \leq 1$.

- Student copula

$$\int_{-\infty}^{\tau_v^{-1}(u)} \int_{-\infty}^{\tau_v^{-1}(v)} \frac{\Gamma(\frac{v+2}{2})}{\Gamma(\frac{v}{2})\pi v \sqrt{1-\theta^2}} \left(1 + \frac{x' \Omega^{-1} x}{\nu} \right)^{-(\nu+2)/2} dx$$

where θ = parameter of copulas, $-1 \leq \theta \leq 1$; t_{ν}^{-1} = quantile function of the Student distribution with ν degrees of freedom; $x = (x_1, x_2)'$; and Ω = correlation matrix as follows:

$$\Omega = \begin{pmatrix} 1 & \theta \\ \theta & 1 \end{pmatrix}$$

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Notation

The following symbols are used in this paper:

- D = flood duration (day);
- d_T = flood duration at T years return period (day);
- $F_D(d)$ = cumulative probability of flood duration D (dimensionless);
- $F_H(h)$ = cumulative probability of flood peak level H (dimensionless);
- H = flood peak level (m);
- h_T = flood peak level at T years return period (m);
- k = shape parameter of GPD (dimensionless);
- k_0 = standard frequency factor used to calculate threshold value μ (dimensionless);
- $M(h, d)$ = joint cumulative probability of flood peak level H and flood duration D (dimensionless);
- N_e = total length of flood time series (dimensionless);
- N_s = total number of sampled extremes (dimensionless);
- r = number of classes for chi-square goodness-of-fit test;
- T_c = time of concentration (day);
- $T_{H,D}$ = joint return period of flood event with flood peak level H and flood duration D (year);
- $T(h|d)$ = conditional return period of event $H > h$ when $D = d$ (year);
- α = scale parameter of GPD (dimensionless);
- θ = parameter of copula;
- μ = threshold value (dimensionless);
- μ_x = mean value (dimensionless);
- ν = degrees of freedom of Student copula (dimensionless);
- σ_x = standard deviation (dimensionless);
- τ = Kendall rank correlation coefficient (dimensionless);
- Φ = Laplace integral; and
- Φ^{-1} = inverse function of Laplace integral.

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